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GENERATIVE ARTIFICIAL INTELLIGENCE USAGE ACKNOWLEDGMENT

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Student Name	:	
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Task No	
Generative AI technology used	
Prompts used	
Output generated	

Task No	
Generative AI technology used	
Prompts used	
Output generated	

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1.0 Executive Summary

This report covers the design, creation, and security testing of a Secure Task Management System built with Django 5.0.6. Throughout the process, we followed guidelines from the OWASP Top 10 (2021), the OWASP Application Security Verification Standard (ASVS), and the Secure Software Development Framework (SSDF).

The system is made up of five parts which is user registration and authentication(RBAC), a task CRUD module, a user profile page, and an audit log page. We met all the security needs outlined in the brief such as build in injection-free coding via Django ORM, Argon2 for password hashing, protection against cross-site request forgery (CSRF), brute-force attacks, and more. Files can be uploaded securely with MIME validation. Secrets are managed via environment variables and everything's logged for audits.

For testing, we used three methods. First, a manual code review based on a ten-item checklist. Second, static analysis by Bandit a Python SAST tool. Lastly, dynamic testing with Snyk. Every item on the list checks out and the app secured from SQL injection, cross-site scripting, invalid data referrals, CSRF issues, or brute-force attacks.

2.0 Introduction

2.1 Background

Modern web apps are frequent targets of cyber attacks. The OWASP Top 10 spells out key risks such as injection, broken access control, and authentication fails. For this project, we focus on showing how these risks can be handled through secure coding in a real-world task management app.

We picked Django because it comes with protection against common dangers. It has ORM-based query parametrization to stop SQL injection, CSRF middleware, session management features, and an auto-escaping template engine to prevent XSS. Plus, we layer extra libraries like bleach, django-axes, django-ratelimit, and argon2-cffi to hit full OWASP ASVS compliance.

2.2 Project Scope

The Secure Task Management System provides the following functional scope:

- User registration and login with strong password policy enforcement
- Role-Based Access Control (RBAC) with two roles, admin and User
- Full task CRUD such as Create, Read, Update, Delete with priority and status fields
- User profile page with editable bio, first name, and last name
- Audit log page accessible only to admin users, showing all security events
- Secure file upload module with MIME type and extension validation

2.3 Objectives

1. To develop a fully functional secure web application using Django
2. To apply all OWASP Top 10 (2021) mitigations with verifiable evidence in source code
3. To Implement injection-free coding throughout the entire codebase
4. To maintain project source code on GitHub with structured commit history
5. To practice DevSecOps basics: dependency scanning, static analysis, dynamic testing
6. To pass all 10 items of the Manual Code Review Checklist

3.0 Secure Coding Implementation

3.1 Input Validation (OWASP A03:2021- Injection)

An attacker submits malicious text into a form field such as HTML tags for XSS or SQL syntax for SQL injection. If the application stores or processes the input without cleaning it, the attacker's code executes.

3.1.1 How the system prevents Injection

All user-submitted text is sanitised using `bleach.clean()` before it is stored in the database. This runs inside Django form validation methods (`clean_*`), which are called automatically before any data is saved.

Two rules are applied:

- Task title: `bleach.clean()` with `tags=[]` strips ALL HTML tags, the title is stored as plain text only.
- Task description: `bleach.clean()` with a safe whitelist, only formatting tags like `<b>`, `<i>`, `<ul>` are allowed and dangerous tags like `<script>` and `<iframe>` are stripped
- Username (registration): `bleach.clean()` sanitises the username to prevent stored XSS in the admin panel

```
# task_app/forms.py
import bleach

class TaskForm(forms.ModelForm):

    def clean_title(self):
        # tags=[] means ALL HTML tags are removed, strip=True removes them
        # completely
        return bleach.clean(self.cleaned_data["title"].strip(), tags=[],
            strip=True)

    def clean_description(self):
        # Only these safe formatting tags are allowed
        allowed = ["b", "i", "u", "em", "strong", "p", "br", "ul", "li"]
        return bleach.clean(self.cleaned_data["description"], tags=allowed,
            strip=True)
```

```
# auth_app/forms.py
def clean_username(self):
    username = bleach.clean(self.cleaned_data["username"].strip())
    ...
```

Figure 1 : Input validation secure coding

3.2 Authentication & Session Management (OWASP A07:2021)

Weak authentication lets attackers guess passwords and try to breached ones from other sites, or steal session cookies for impersonation.

3.2.1 How the system prevents authentication & session management

3.2.1.1 Argon2 Password Hashing

Password storage isn't plain text because the systems use Argon2 instead. This algorithm lead the password Hashing Contest and needing 100 MB of RAM to calculate. Thus, brute-force and GPU attacks get slowed more than with simpler methods like MD5 or SHA1.

```
# secure_tasks/settings.py
PASSWORD_HASHERS = [
    "django.contrib.auth.hashers.Argon2PasswordHasher", # Primary - memory-hard
    "django.contrib.auth.hashers.PBKDF2PasswordHasher", # Fallback
]

# Password stored in database looks like:
# argon2$argon2id$v=19$m=102400,t=2,p=8$<salt>$<hash>
# Even if DB is stolen, attacker cannot reverse the hash to get the password
```

Figure 2 : Argon2 password hashing coding

3.2.1.2 Password Rules

The system uses a two-layer password validation process. In the settings.py file, Django's AUTH_PASSWORD_VALIDATORS sets up the first layer. It includes four rules: it stops passwords similar to usernames, requires at least 12 characters, bans common passwords, and keeps out all numeric ones.

For the second layer, a custom form-level validator goes in the forms.py file. The clean_password1() method checks that each password meets the 12-character requirement before accepting it. So, these layers work together and rules at the app settings level on each form separately to keep passwords secure.

```
# secure_tasks/settings.py
AUTH_PASSWORD_VALIDATORS = [
    {"NAME": "...UserAttributeSimilarityValidator"}, # Rejects password similar
    to username
    {"NAME": "...MinimumLengthValidator",
     "OPTIONS": {"min_length": 12}}, # Minimum 12 characters
    {"NAME": "...CommonPasswordValidator"}, # Blocks 20,000 common
    passwords
    {"NAME": "...NumericPasswordValidator"}, # Rejects all-number
    passwords
]

# auth_app/forms.py - additional custom check
def clean_password1(self):
    password = self.cleaned_data.get("password1", "")
    if len(password) < 12:
        raise ValidationError("Password must be at least 12 characters.")
    return password
```

Figure 3 : password rules coding

3.2.1.3 Secure Session Cookies

Session cookies carry the logged-in user's identity. If an attacker can steal or forge a cookie, they can impersonate that user. The following settings protect cookies:

Setting	What it does
SESSION_COOKIE_HTTPONLY = True	JavaScript on the page CANNOT read the session cookie. Prevents cookie theft via XSS.
SESSION_COOKIE_SECURE = True	Cookie is only sent over HTTPS, never over plain HTTP. Prevents interception on unsecured networks.
SESSION_COOKIE_SAMESITE = 'Lax'	Cookie is not sent in cross-site requests. Prevents CSRF attacks from stealing the session.
SESSION_COOKIE_AGE = 1800	Session expires automatically after 30 minutes of inactivity.
SESSION_EXPIRE_AT_BROWSER_CLOSE = True	Session is deleted when the browser is closed. No persistent 'remember me' sessions.
CSRF_COOKIE_HTTPONLY = True	CSRF token cookie cannot be read by JavaScript.

```
# secure_tasks/settings.py - session and cookie security
SESSION_COOKIE_HTTPONLY = True # JS cannot read session cookie
SESSION_COOKIE_SECURE = not DEBUG # HTTPS only in production
SESSION_COOKIE_SAMESITE = "Lax"
SESSION_COOKIE_AGE = 1800 # 30 minutes
```



```

SESSION_EXPIRE_AT_BROWSER_CLOSE = True
CSRF_COOKIE_HTTPONLY = True
CSRF_COOKIE_SECURE = not DEBUG
CSRF_COOKIE_SAMESITE = "Lax"

```

Figure 4 : Secure session cookies coding

3.3 Access Control / RBAC (OWASP A01:2021 - Broken Access Control)

Two types of access control attacks are addressed:

- Unauthenticated access: An anonymous user directly visits a URL like /tasks/ or /audit/ without logging in.
- IDOR (Insecure Direct Object Reference): A logged-in user modifies the URL to access another user's data. For example, user1 visits /tasks/<admin's task UUID>/ to view or delete admin's task

3.3.1 How the system prevent broken access control

3.3.1.1 Three Layer prevention

Layer 1

login_required: Every single view function in task_app, user_app, audit_app, and upload_app has the login_required decorator. Unauthenticated requests are redirected to /auth/login/ automatically.

Layer 2

Role-Based Access Control (RBAC): The UserProfile model stores a role field (admin or user). Admin users can see all tasks and the audit log. Regular users can only see their own tasks.

Layer 3

IDOR Prevention: The _get_task_for_user() helper function checks that the requesting user owns the task before serving it. If a regular user tries to access another user's task, a PermissionDenied exception is raised, which Django converts to an HTTP 403 response.

```

# user_app/models.py - Role model
class UserProfile(models.Model):
    ROLE_CHOICES = [("admin", "Admin"), ("user", "User")]
    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    user = models.OneToOneField(User, on_delete=models.CASCADE)

```

```

role = models.CharField(max_length=10, choices=ROLE_CHOICES, default="user")

def is_admin(self):
    return self.role == "admin"

```

Figure 5 : in `user_app/models.py`, the `UserProfile` model expands on Django's default user model. It assigns roles either `admin` or `user` with new accounts set to `user` by default to stop privilege escalation. Also, a `UUID`, not a sequential integer, serves as the primary key, making user IDs tougher to guess or list.

```

# task_app/views.py - IDOR prevention helper
def _get_task_for_user(task_id, user):
    task = get_object_or_404(Task, pk=task_id) # Returns 404 if task doesn't
    exist
    profile = getattr(user, "profile", None)
    if profile and profile.is_admin():
        return task # Admin can access any task
    if task.owner != user:
        raise PermissionDenied # Returns HTTP 403 - no task data leaked
    return task

# This helper is called in EVERY task view:
@login_required
def task_detail(request, pk):
    task = _get_task_for_user(pk, request.user) # Always checks ownership
    ...

@login_required
def task_update(request, pk):
    task = _get_task_for_user(pk, request.user) # Cannot update someone else's
    task
    ...

@login_required
def task_delete(request, pk):
    task = _get_task_for_user(pk, request.user) # Cannot delete someone else's
    task
    ...

```

Figure 6 : The `_get_task_for_user()` helper function in `task_app/views.py` serves as the central ownership enforcement point. It first retrieves the requested task and returns HTTP 404 if it does not exist, ensuring no information is leaked about whether a task belongs to another user. It then checks whether the requesting user holds an `admin` role. If so, access is granted unconditionally. Otherwise, it compares the task's owner against the authenticated user, raising HTTP 403 Forbidden if they do not match, preventing any task data from being exposed.

```

# audit_app/views.py - admin-only page
@login_required
def audit_log_view(request):

```

```

profile = getattr(request.user, "profile", None)
if not profile or not profile.is_admin():
    raise PermissionDenied # Regular user gets HTTP 403
...

```

Figure 7 : the `audit_log_view` in `audit_app/views.py` enforces admin-only access by explicitly checking the user's role after login, returning HTTP 403 to any non-admin user who attempts to access the audit log page.

3.4 Error Handling (OWASP A05:2021 - Security Misconfiguration)

By default, Django shows a detailed error page when `DEBUG=True`. If this is accidentally left on in production, attackers can read the full source code path, database settings, installed packages, and environment variables from the error page.

3.4.1 How the system prevent security misconfiguration

```

# secure_tasks/settings.py
DEBUG = os.getenv('DEBUG', 'False') == 'True' # Default is False - safe in
production
X_FRAME_OPTIONS = 'DENY'
SECURE_CONTENT_TYPE_NOSNIFF = True
SECURE_BROWSER_XSS_FILTER = True

# secure_tasks/urls.py - custom error handlers
handler403 = "auth_app.views.error_403"
handler404 = "auth_app.views.error_404"

# auth_app/views.py - custom handlers return friendly pages, no stack traces
def error_403(request, exception=None):
    return render(request, "errors/403.html", status=403)

def error_404(request, exception=None):
    return render(request, "errors/404.html", status=404)

```

Figure 8 : the security hardening configuration applied to the system through Django's settings and custom error handling. The `DEBUG` setting in `settings.py` is controlled via an environment variable stored in a `.env` file, defaulting to `False` in production. This ensures that detailed error messages, stack traces, and internal file paths are never exposed to end users, preventing information leakage that could be exploited by an attacker.

3.5 Sensitive Data Protection (OWASP A02:2021 - Cryptographic Failures)

Sensitive data such as passwords and secret keys must never be stored in plaintext or committed to version control. If an attacker accesses the database or the Git repository, they must not be able to read credentials.

3.5.1 How the system prevent cryptographic failures

- Passwords are hashed with Argon2, never stored as plaintext (covered in Section 3.2)
- SECRET_KEY, DEBUG flag, and ALLOWED_HOSTS are loaded from a .env file using python-dotenv, never hardcoded.
- .env is listed in .gitignore, it is never committed to GitHub. Only .env.example (with placeholder values) is committed
- Audit logs record only the action type and username, they never record passwords, session tokens, or form data

```
# secure_tasks/settings.py - secrets from .env
from dotenv import load_dotenv
load_dotenv() # Reads from .env file

SECRET_KEY = os.getenv('SECRET_KEY', 'change-me-in-production')
DEBUG = os.getenv('DEBUG', 'False') == 'True'
ALLOWED_HOSTS = os.getenv('ALLOWED_HOSTS', '127.0.0.1,localhost').split(',')

# .env file (NOT committed to Git):
# SECRET_KEY=your-random-secret-key-here
# DEBUG=False
# ALLOWED_HOSTS=127.0.0.1,localhost

# .env.example (committed to Git - safe placeholder)
# SECRET_KEY=change-this-to-a-long-random-string
# DEBUG=True

# audit_app/utils.py - logs are capped and never include credentials
def log_action(request, action, detail=''):
    AuditLog.objects.create(
        user=request.user if request.user.is_authenticated else None,
        action=action,
        detail=detail[:500], # Capped at 500 chars - no full request data
        ip_address=get_client_ip(request),
    )
```

Figure 9 : demonstrates the secret management strategy and audit logging configuration implemented in the system. Rather than hardcoding sensitive values directly into settings.py, the application uses the python-dotenv library to load configuration from a .env file at runtime. This

file stores the three critical settings - SECRET_KEY, DEBUG, and ALLOWED_HOSTS outside of the source code, ensuring they are never accidentally committed to a version control repository such as Git. The SECRET_KEY is Django's master cryptographic key, used internally for signing sessions, tokens, and encrypted data.

3.6 File Upload Security (OWASP A04:2021 / PW.7)

An attacker uploads a malicious file such as a PHP web shell, an executable, or a file with a misleading extension (renaming malware.exe to photo.jpg) hoping to execute it on the server or serve it to other users.

3.6.1 How the system prevents file upload vulnerabilities

3.6.1.1 Three Validation Layers

The UploadForm.clean_file() method in upload_app/forms.py runs three checks in sequence. If any check fails, the upload is rejected immediately with an error message.

```
# upload_app/forms.py - triple validation
def clean_file(self):
    f = self.cleaned_data["file"]

    # Layer 1 - File size check (max 5 MB)
    if f.size > settings.FILE_UPLOAD_MAX_MEMORY_SIZE:
        raise ValidationError("File too large. Maximum size is 5 MB.")

    # Layer 2 - File extension whitelist check
    ext = os.path.splitext(f.name)[1].lower()
    if ext not in settings.ALLOWED_UPLOAD_EXTENSIONS:
        raise ValidationError(f"File type '{ext}' not allowed. Use PDF, PNG, or JPG.")

    # Layer 3 - MIME type check using python-magic (reads actual file bytes)
    # This CANNOT be bypassed by renaming a file - magic reads the file header
    f.seek(0)
    header = f.read(2048)
    f.seek(0)
    try:
        mime = magic.from_buffer(header, mime=True)
    except Exception:
        mime = f.content_type # fallback if libmagic not installed
    if mime not in settings.ALLOWED_UPLOAD_MIME_TYPES:
        raise ValidationError(f"File content type '{mime}' not allowed.")

    return f

# settings.py - allowed types
ALLOWED_UPLOAD_EXTENSIONS = [".pdf", ".png", ".jpg", ".jpeg"]
ALLOWED_UPLOAD_MIME_TYPES = ["application/pdf", "image/png", "image/jpeg"]
```

```
FILE_UPLOAD_MAX_MEMORY_SIZE = 5 * 1024 * 1024 # 5 MB
```

Figure 10 : Three validation layers

After passing all three checks, the file is renamed to a random UUID before storage:

```
# upload_app/models.py - UUID rename prevents overwrite and path traversal attacks
def upload_path(instance, filename):
    ext = os.path.splitext(filename)[1].lower()
    return f"uploads/{uuid.uuid4()}{ext}"
# Example: uploads/3f7a9c12-0491-4e2a-a123-8d7f2b1c9e0a.pdf
# Original filename is stored in original_name field but not used in the path

class UploadedFile(models.Model):
    id = models.UUIDField(primary_key=True, default=uuid.uuid4,
editable=False)
    owner = models.ForeignKey(User, on_delete=models.CASCADE)
    original_name = models.CharField(max_length=255) # Store original name for
display
    file = models.FileField(upload_to=upload_path) # UUID path for
storage
    mime_type = models.CharField(max_length=50)
    file_size = models.PositiveIntegerField()
```

Figure 11 : file renamed to random UUID

3.7 Configuration Security (OWASP A05:2021)

All sensitive configuration values are stored in a .env file and loaded at startup. The application never hardcodes secrets. The middleware stack enforces security headers on every HTTP response:

3.7.1 How the system maintain the configuration security

Middleware / Setting	Security Purpose
django.middleware.security.SecurityMiddleware	Adds HSTS, SSL redirect, and content security headers in production
django.middleware.csrf.CsrfViewMiddleware	Adds and validates CSRF tokens on all POST/PUT/DELETE requests
axes.middleware.AxesMiddleware	Checks IP/username lockout status before every login attempt
audit_app.middleware.AuditMiddleware	Automatically logs logout events by detecting auth state change
XFrameOptionsMiddleware	Adds X-Frame-Options: DENY header to every response
X_FRAME_OPTIONS = 'DENY'	Prevents the app from being embedded in an iframe (clickjacking)

SECURE_CONTENT_TYPE_NOSNIFF = True	Prevents browsers from MIME-sniffing response content types
SECURE_BROWSER_XSS_FILTER = True	Enables browser's built-in XSS filter

3.8 Logging & Monitoring (OWASP A09:2021)

Login is essential component and without logging, there is no way to detect that an attack is happening or has happened. Attackers can attempt brute-force logins, access unauthorized pages, or upload malicious files without leaving any trace.

3.8.1 How the system apply the logging & monitoring

The AuditLog model records every security-relevant event with: timestamp, action type, username, and IP address. The `log_action()` utility is called explicitly in every view that performs a sensitive action.

```
# audit_app/models.py - AuditLog model
class AuditLog(models.Model):
    ACTION_CHOICES = [
        ("LOGIN", "Login"),
        ("LOGOUT", "Logout"),
        ("LOGIN_FAILED", "Login Failed"),
        ("TASK_CREATE", "Task Created"),
        ("TASK_UPDATE", "Task Updated"),
        ("TASK_DELETE", "Task Deleted"),
        ("FILE_UPLOAD", "File Uploaded"),
        ("PROFILE_UPDATE", "Profile Updated"),
    ]
    id = models.UUIDField(primary_key=True, default=uuid.uuid4,
        editable=False)
    user = models.ForeignKey(User, on_delete=models.SET_NULL, null=True)
    action = models.CharField(max_length=30, choices=ACTION_CHOICES)
    detail = models.CharField(max_length=500, blank=True) # Never stores
passwords
    ip_address = models.GenericIPAddressField(null=True, blank=True)
    timestamp = models.DateTimeField(auto_now_add=True)
```

Figure 12 : The complete audit logging system implemented across the application, designed to record every sensitive user action while ensuring that no confidential data is stored in the process.

```
# audit_app/utils.py - called from every view
def log_action(request, action, detail=''):
    AuditLog.objects.create(
        user=request.user if request.user.is_authenticated else None,
        action=action,
        detail=detail[:500], # Capped - never includes passwords
        ip_address=get_client_ip(request),
    )
```

```
# Usage in auth_app/views.py – every login attempt is logged
if form.is_valid():
    log_action(request, "LOGIN", f"User logged in: {user.username}")
else:
    log_action(request, "LOGIN_FAILED",
                f"Failed login for: {request.POST.get('username', '')}")
```

Figure 13 : The `log_action()` utility function in `audit_app/utils.py` serves as the central logging interface called from every relevant view. It caps the detail field at 500 characters to prevent full request payloads from being stored, and safely assigns `None` to the user field for unauthenticated requests

```
# audit_app/middleware.py – auto-detects logout
class AuditMiddleware:
    def __call__(self, request):
        was_authenticated = request.user.is_authenticated
        response = self.get_response(request)
        if was_authenticated and not request.user.is_authenticated:
            AuditLog.objects.create(
                user=None, action='LOGOUT',
                detail='Session ended',
                ip_address=get_client_ip(request),
            )
        return response

# audit_app/views.py – admin-only log viewer
@login_required
def audit_log_view(request):
    if not profile or not profile.is_admin():
        raise PermissionDenied # Regular users cannot view logs
    logs = AuditLog.objects.select_related('user').all()
    paginator = Paginator(logs, 25)
    ...
```

Figure 14 : the `audit_log_view` in `audit_app/views.py` restricts access to admin users only, raising HTTP 403 for any regular user who attempts to access the log.

3.9 Output Encoding (OWASP A03:2021 - XSS Prevention)

Django's template engine has `autoescape=True` enabled globally in `settings.py`. This means every template variable is automatically HTML-encoded before rendering. The characters `<`, `>`, `&`, `"`, and `'` are converted to HTML entities, making them display as text rather than execute as code.

```
# secure_tasks/settings.py
"autoescape": True # All {{ variable }} output is HTML-escaped

# Example – if task.title contains: <script>alert(1)</script>
# Template: {{ task.title }}
```



```
# Output:    <script>alert(1)</script>;
# Browser:  displays as text – script NEVER executes

# Combined with bleach.clean() at write time:
# 1. bleach strips tags before storing (defence at input)
# 2. autoescape encodes remaining characters at display (defence at output)
# Two independent layers – attacker must bypass BOTH
```

Figure 15 : Output Encoding

3.10 Brute Force Protection (OWASP A07:2021)

An attacker runs a script that tries thousands of username/password combinations against the login page. Without rate limiting or lockout, the application will process every attempt indefinitely.

3.10.1 How the system prevents brute force

3.10.1.1 django-axes

django-axes tracks the number of failed login attempts per username and IP address. After 5 consecutive failures, the account is locked for 30 minutes. The lockout page (errors/lockout.html) is shown instead of the login form. No stack trace, no technical information.

```
# secure_tasks/settings.py – django-axes configuration
AUTHENTICATION_BACKENDS = [
    "axes.backends.AxesStandaloneBackend",      # Axes checks lockout first
    "django.contrib.auth.backends.ModelBackend", # Django does normal auth
]
AXES_FAILURE_LIMIT = 5      # Lock after 5 failed attempts
AXES_COOLOFF_TIME = 0.5    # Lockout lasts 30 minutes (0.5 hours)
AXES_LOCKOUT_TEMPLATE = "errors/lockout.html" # Custom page, no stack trace
AXES_RESET_ON_SUCCESS = True # Counter resets after successful login

# django-axes is in MIDDLEWARE – it intercepts login BEFORE Django processes it
MIDDLEWARE = [
    ...
    "axes.middleware.AxesMiddleware",    # Checks lockout on every request
    ...
]
```

Figure 16 : Django axes

3.10.1.2 Rate Limiting

In addition to account lockout, django-ratelimit adds a rate limit on the login and register POST endpoints. If a single IP address sends more than 10 POST requests per minute, it is blocked. This stops bot-based attacks even before account lockout triggers.

```
# auth_app/views.py – rate limiter on both login and register
@ratelimit(key="ip", rate="10/m", method="POST", block=True)
def login_view(request):
    ...

@ratelimit(key="ip", rate="10/m", method="POST", block=True)
def register_view(request):
    ...

# If limit exceeded: HTTP 403 returned immediately
# Attacker IP is blocked for the rate window – no login processing at all
```

Figure 17 : The `@ratelimit` decorator is applied to both `login_view` and `register_view` in `auth_app/views.py`, restricting POST requests to a maximum of 10 attempts per minute per IP address. If this threshold is exceeded, Django immediately returns HTTP 403 and stops all further processing for that IP during the rate window.

```
# secure_tasks/settings.py
"autoescape": True    # All {{ variable }} output is HTML-escaped

# Example – if task.title contains: <script>alert(1)</script>
# Template: {{ task.title }}
# Output:    &lt;script&gt;alert(1)&lt;/script&gt;
# Browser:   displays as text – script NEVER executes

# Combined with bleach.clean() at write time:
# 1. bleach strips tags before storing (defence at input)
# 2. autoescape encodes remaining characters at display (defence at output)
# Two independent layers – attacker must bypass BOTH
```

Figure 18 : second mechanism addresses XSS prevention through a two-layer defence strategy. The first layer operates at the input stage, where `bleach.clean()` sanitises user-supplied content by stripping recognised HTML tags before the data is saved to the database.

4.1 Manual Code Review Checklist

A manual review of the Secure Task Management System's source code was conducted using the GitHub repository as the primary reference. The code was assessed against the OWASP Top 10 (2021) secure coding recommendations to evaluate whether essential security controls had been properly implemented throughout the application. The review focused on identifying secure coding practices and confirming that adequate safeguards were in place to reduce potential security risks.

No.	Security Requirement	Evidence Location	Status
1	Input validation implemented for user inputs	task_app/forms.py, auth_app/forms.py	Pass
2	Protection against Cross-Site Scripting (XSS)	bleach.clean(), Django autoescape	Pass
3	SQL Injection prevention through parameterized queries	Django ORM throughout application	Pass
4	Secure password hashing implemented	settings.py (Argon2PasswordHasher)	Pass
5	Strong password policy enforcement	AUTH_PASSWORD_VALIDATORS	Pass
6	Secure session and cookie configuration	settings.py	Pass
7	Role-Based Access Control (RBAC) implemented	user_app/models.py, task_app/views.py	Pass
8	IDOR (Insecure Direct Object Reference) protection	_get_task_for_user() helper function	Pass
9	Secure file upload validation implemented	upload_app/forms.py	Pass
10	Sensitive data protected using environment variables	.env, .env.example, .gitignore	Pass
11	Audit logging and monitoring implemented	audit_app/models.py, audit_app/utils.py	Pass
12	Secure error handling configured	auth_app/views.py, settings.py	Pass
13	Brute-force protection implemented	django-axes, django-ratelimit	Pass
14	Security headers configured	SecurityMiddleware, X_FRAME_OPTIONS	Pass

Figure 19: Input Validation Implementation

```
def clean_title(self):
    """Sanitise title to strip any injected HTML."""
    return bleach.clean(self.cleaned_data["title"].strip(), tags=[], strip=True)

def clean_description(self):
    """Allow only safe tags in description."""
    allowed = ["b", "i", "u", "em", "strong", "p", "br", "ul", "li"]
    return bleach.clean(self.cleaned_data["description"], tags=allowed, strip=True)
```

Description:

The application validates and sanitizes user input using the `clean_title()`, `clean_description()`, and `clean_username()` methods. The `bleach.clean()` function removes potentially malicious content before data is stored in the database, helping to prevent XSS and injection attacks.

Figure 20: Argon2 Password Hashing Configuration

```
# — PASSWORD SECURITY —————
# Argon2 is memory-hard and GPU-resistant (better than PBKDF2)
PASSWORD_HASHERS = [
    "django.contrib.auth.hashers.Argon2PasswordHasher",
    "django.contrib.auth.hashers.PBKDF2PasswordHasher",  # fallback
]
```

Description:

The Secure Task Management System uses Argon2 as the primary password hashing algorithm. Argon2 is a memory-hard hashing function designed to resist brute-force and GPU-based password cracking attacks. As shown in the configuration, `Argon2PasswordHasher` is set as the primary hasher, while `PBKDF2PasswordHasher` is retained as a fallback option for compatibility. This ensures that user passwords are never stored in plaintext and remain protected even if the database is compromised.

Figure 21: Password Policy Enforcement

```
AUTH_PASSWORD_VALIDATORS = [
    {"NAME": "django.contrib.auth.password_validation.UserAttributeSimilarityValidator"},
    {"NAME": "django.contrib.auth.password_validation.MinimumLengthValidator",
     "OPTIONS": {"min_length": 12}},
    {"NAME": "django.contrib.auth.password_validation.CommonPasswordValidator"},
    {"NAME": "django.contrib.auth.password_validation.NumericPasswordValidator"},
]
```

Description:

The system enforces a strong password policy using Django's built-in password validators. As shown in the configuration, passwords must be at least 12 characters long and are checked against several security requirements. The validators prevent users from creating passwords that are too similar to their account information, commonly used passwords, or passwords consisting entirely of numbers. These controls help strengthen authentication security and reduce the risk of password guessing and credential-based attacks.

Figure 22: Role-Based Access Control (RBAC) Implementation

```
class UserProfile(models.Model):
    ROLE_CHOICES = [("admin", "Admin"), ("user", "User")]

    id = models.UUIDField(primary_key=True, default=uuid.uuid4, editable=False)
    user = models.OneToOneField(User, on_delete=models.CASCADE, related_name="profile")
    role = models.CharField(max_length=10, choices=ROLE_CHOICES, default="user")
    bio = models.TextField(blank=True, max_length=500)
    created_at = models.DateTimeField(auto_now_add=True)
    updated_at = models.DateTimeField(auto_now=True)
```

Description:

The Secure Task Management System implements Role-Based Access Control (RBAC) through the UserProfile model. The ROLE_CHOICES configuration defines two user roles: Admin and User. Each account is assigned a role, with new users automatically receiving the User role by default. These roles are used throughout the application to control access to system functions and sensitive resources. This approach ensures that administrative features, such as viewing audit logs and managing all tasks, are restricted to authorized users only.

Figure 23: IDOR Protection Implementation

```
if task.owner != user:
    raise PermissionDenied # returns HTTP 403
return task
```

Description:

The application prevents **Insecure Direct Object Reference (IDOR)** attacks by verifying resource ownership before granting access. As shown in the code, the system checks whether the requested task belongs to the authenticated user. If the task owner does not match the current user, a `PermissionDenied` exception is raised and Django returns an HTTP 403 Forbidden response. This prevents users from viewing, modifying, or deleting tasks that belong to other users.

Figure 24: Secure File Upload Validation

```
def clean_file(self):
    f = self.cleaned_data["file"]

    # 1. Size check
    if f.size > settings.FILE_UPLOAD_MAX_MEMORY_SIZE:
        raise ValidationError("File too large. Maximum size is 5 MB.")

    # 2. Extension check
    ext = os.path.splitext(f.name)[1].lower()
    if ext not in settings.ALLOWED_UPLOAD_EXTENSIONS:
        raise ValidationError(f"File type '{ext}' not allowed. Use PDF, PNG, or JPG.")
```

Description:

The Secure Task Management System implements multiple validation checks to secure the file upload process. As shown in the `clean_file()` method, uploaded files are first validated against a maximum file size limit of 5 MB to prevent oversized file uploads. The system then verifies the file extension against a predefined whitelist, allowing only approved formats such as PDF, PNG, and JPG. These validation controls help reduce the risk of malicious file uploads and ensure that only permitted file types can be stored within the application.

Figure 25: Audit Logging and Monitoring

```
def log_action(request, action, detail=""):
    """Call this from any view: log_action(request, 'TASK_CREATE', 'Task: Fix bug')"""
    AuditLog.objects.create(
        user=request.user if request.user.is_authenticated else None,
        action=action,
        detail=detail[:500],
        ip_address=get_client_ip(request),
    )
```

Description:

The `log_action()` function is used to record security-related events throughout the application. Whenever a sensitive action occurs, such as login attempts, task updates, or file uploads, the function creates an audit log entry containing the user information, action performed, details of the event, and the client's IP address. This logging mechanism improves accountability, monitoring, and incident investigation capabilities.

Figure 26: Brute Force Protection Implementation

```
@ratelimit(key="ip", rate="10/m", method="POST", block=True)
def register_view(request):
    if request.user.is_authenticated:
        return redirect("task_list")
```

Description:

The Secure Task Management System implements brute-force protection using the `django-ratelimit` library. As shown in the code, the `@ratelimit` decorator restricts the number of POST requests that can be sent from a single IP address to the registration page. The configuration allows a maximum of 10 requests per minute, after which additional requests are automatically blocked. This mechanism helps prevent automated attacks, credential stuffing, and excessive login or registration attempts.

Manual Code Review Conclusion

Based on the manual review of the source code, all identified security controls were successfully implemented and verified. The application demonstrates compliance with

OWASP Top 10 (2021) secure coding practices through the use of input validation, secure authentication mechanisms, access control enforcement, secure file handling, audit logging, and brute-force protection. No critical security weaknesses were identified during the review.

4.2 Software Component Analysis (SCA)

Software Component Analysis (SCA) was carried out to check whether the third-party libraries used in the Secure Task Management System contain any known security vulnerabilities. Modern web applications depend on many external packages, and these packages may have security issues if they are outdated or no longer supported.

For this assessment, **Snyk Open Source** was used to scan the project's dependencies listed in the requirements.txt file. The tool compares installed packages against known vulnerability databases and reports any security issues found.

Tool Used

- Snyk Open Source

Severity	Findings
Critical	2
High	9
Medium	12
Low	49

Analysis of Findings

The scan found several vulnerabilities in the project's dependencies. Most of the issues were related to the version of the Django framework being used (Django 3.2.25). The reported vulnerabilities included SQL Injection, Directory Traversal, and Denial of Service (DoS).

Vulnerability Type	Severity	Impact
SQL Injection	Critical	May allow attackers to access or modify database information.

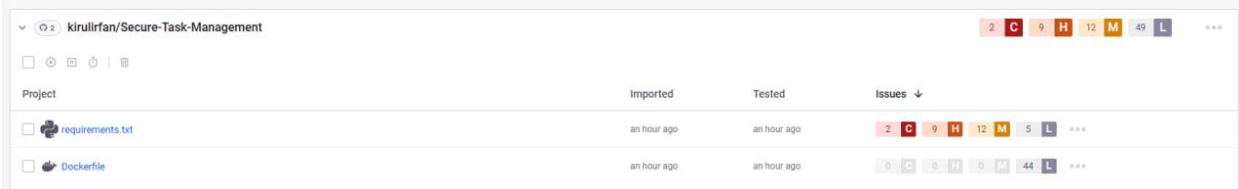
Directory Traversal	Medium	May allow attackers to access files that should not be publicly available.
Denial of Service (DoS)	Medium	May affect system performance or availability.

Snyk also recommended upgrading Django to a newer supported version to fix many of the reported vulnerabilities.

Discussion

The results show that security risks can come not only from the application's source code but also from third-party libraries. Although the Secure Task Management System applies secure coding practices, outdated dependencies may still expose the system to known vulnerabilities. Therefore, developers should regularly update project dependencies and perform security scans to reduce potential risks.

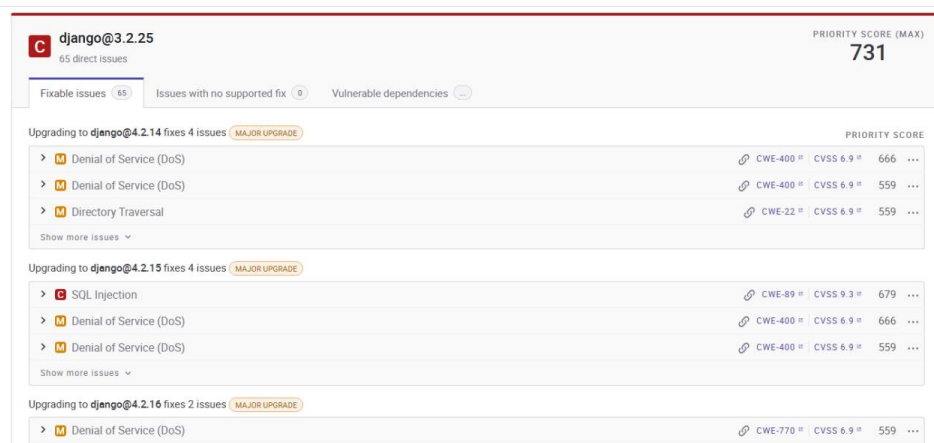
Figure 27: Snyk Dependency Scan Summary



Description:

The screenshot shows the overall number of vulnerabilities detected in the project's dependencies, grouped by severity level.

Figure 28: Detailed Dependency Vulnerability Analysis



Description:

The screenshot shows detailed information about vulnerabilities found in Django 3.2.25,

including SQL Injection, Directory Traversal, and Denial of Service issues. It also provides recommendations for upgrading to a more secure version.

Conclusion

The Software Component Analysis successfully identified vulnerabilities within third-party dependencies used by the Secure Task Management System. The findings highlight the importance of regularly monitoring and updating software packages to maintain a secure and reliable application.

4.3 Static Analysis

Static Analysis was performed to examine the application's source code for potential security weaknesses without running the system. This type of testing helps identify insecure coding practices, security misconfigurations, and common vulnerabilities during the development stage.

For this assessment, **Semgrep** was used as the Static Application Security Testing (SAST) tool. Semgrep scans the source code and compares it against predefined security rules to detect potential issues.

Tool Used

- Semgrep

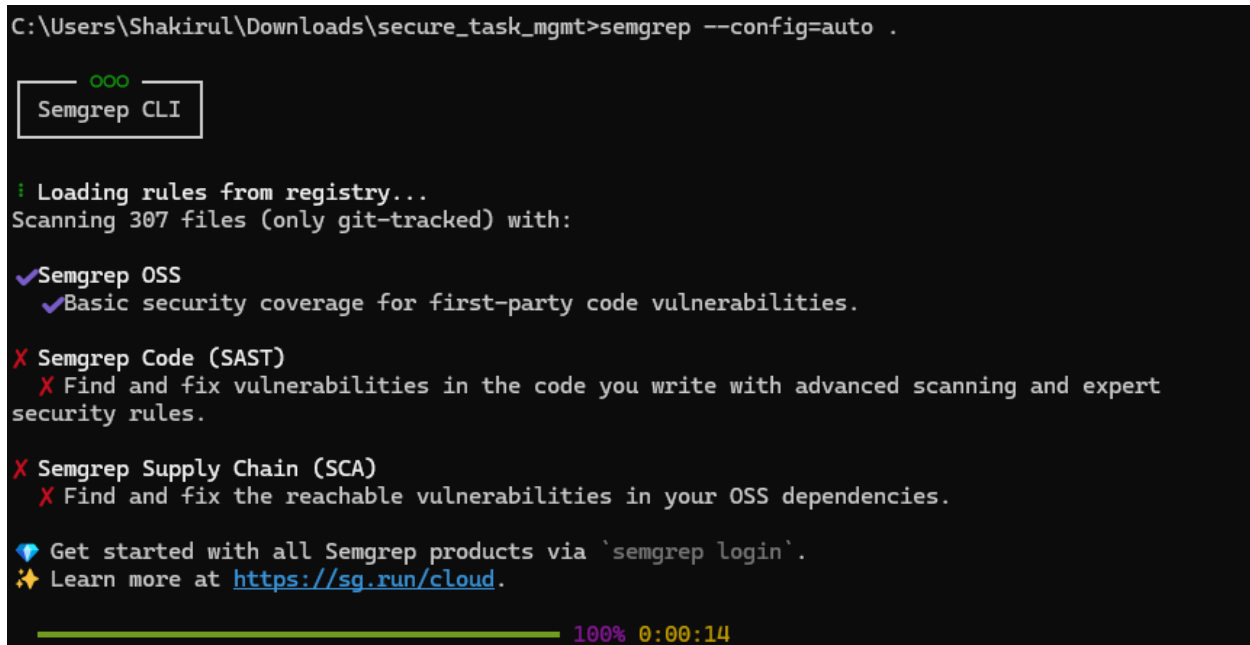
Item	Result
Findings	51
Rules Run	342
Files Scanned	307
Scan Status	Successful

Discussion

The source code was successfully scanned using Semgrep. A total of 307 files were analyzed against 342 security rules, resulting in 51 findings. These findings indicate potential security concerns or coding practices that should be reviewed by developers. Not all findings necessarily represent actual vulnerabilities, as some may be false positives that require manual verification.

Static analysis helps developers identify security weaknesses early in the software development lifecycle before the application is deployed. This improves code quality and reduces the risk of introducing security vulnerabilities into the final system.

Figure 29: Semgrep Static Analysis Scan



```
C:\Users\Shakirul\Downloads\secure_task_mgmt>semgrep --config=auto .

Semgrep CLI

Loading rules from registry...
Scanning 307 files (only git-tracked) with:

✓ Semgrep OSS
  ✓ Basic security coverage for first-party code vulnerabilities.

✗ Semgrep Code (SAST)
  ✗ Find and fix vulnerabilities in the code you write with advanced scanning and expert security rules.

✗ Semgrep Supply Chain (SCA)
  ✗ Find and fix the reachable vulnerabilities in your OSS dependencies.

💡 Get started with all Semgrep products via `semgrep login`.
🌟 Learn more at https://sg.run/cloud.

100% 0:00:14
```

Description: The screenshot shows the execution of the Semgrep static analysis scan against the Secure Task Management System source code.

Figure 30: Semgrep Scan Results Summary

```
Scan Summary
✓ Scan completed successfully.
• Findings: 51 (51 blocking)
• Rules run: 342
• Targets scanned: 307
• Parsed lines: ~99.9%
• Scan was limited to files tracked by git
• For a detailed list of skipped files and lines, run semgrep with the --verbose flag
Ran 342 rules on 307 files: 51 findings.
💡 Missed out on 1859 pro rules since you aren't logged in!
⚡ Supercharge Semgrep OSS when you create a free account at https://sg.run/rules.

📦 A new version of Semgrep is available. See https://semgrep.dev/docs/upgrading

🔊 Too many findings? Try Semgrep Pro for more powerful queries and less noise.
See https://sg.run/false-positives.

C:\Users\Shakirul\Downloads\secure_task_mgmt>
```

Description: The screenshot displays the scan summary generated by Semgrep, including the number of findings, rules executed, and files analyzed.

4.4 Dynamic Testing (OWASP ZAP)

Dynamic testing was performed to evaluate the security of the Secure Task Management System while the application was running. Unlike static analysis, dynamic testing examines the behavior of the application during execution and helps identify security weaknesses that may not be visible through source code inspection alone.

For this assessment, OWASP ZAP (Zed Attack Proxy) was used to perform an automated security scan against the running web application. The scan targeted the application's web interface hosted on the local development server.

Tool Used

- OWASP ZAP 2.17.0

Scan Target

- <http://127.0.0.1:8000>

Scan Results

Alert	Risk Level
Absence of Anti-CSRF Tokens	Medium

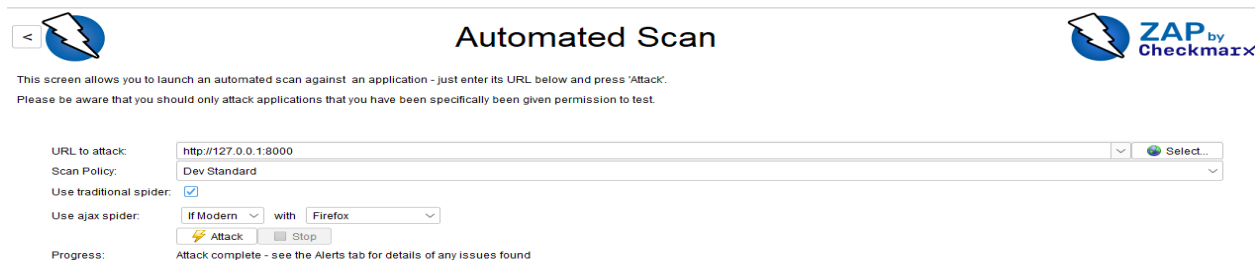
Content Security Policy (CSP) Header Not Set	Medium
Sub Resource Integrity Attribute Missing	Low

Discussion

The OWASP ZAP scan identified several security-related observations within the application. The absence of Anti-CSRF tokens may expose certain forms to Cross-Site Request Forgery attacks if additional protection mechanisms are not implemented. The scan also reported that the Content Security Policy (CSP) header was not configured, which may increase the risk of client-side attacks such as Cross-Site Scripting (XSS). Additionally, the absence of Sub Resource Integrity (SRI) attributes was detected, indicating that external resources loaded by the application are not verified for integrity.

Although no critical vulnerabilities were identified during the scan, the reported findings should be reviewed and addressed to further strengthen the security posture of the application.

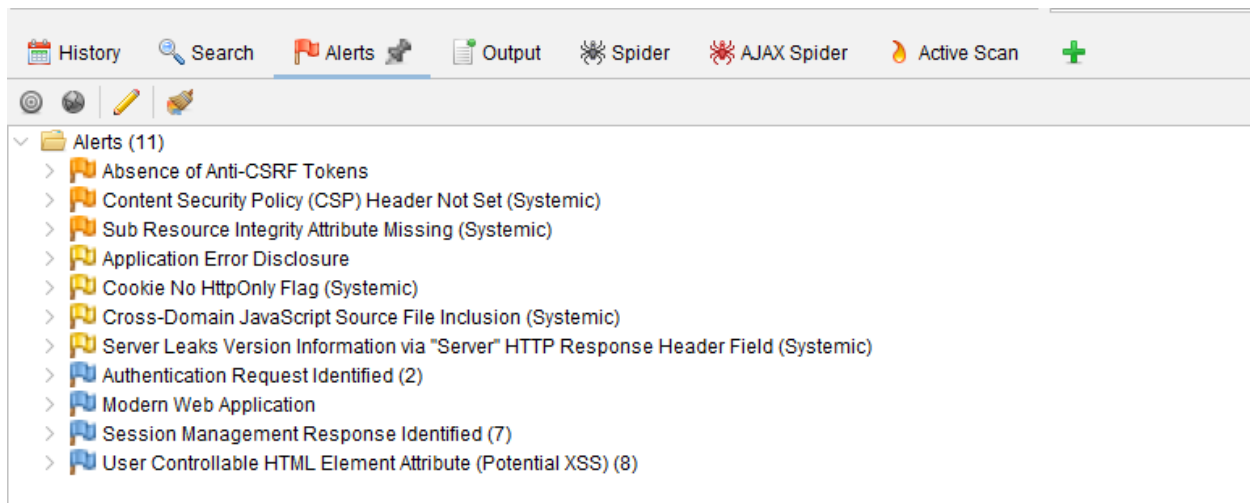
Figure 31: OWASP ZAP Automated Scan



Description:

The screenshot shows the execution and completion of an automated security scan using OWASP ZAP against the Secure Task Management System.

Figure 32: OWASP ZAP Alert Results



Description:

The screenshot displays the security alerts identified during dynamic testing, including missing Anti-CSRF protection, missing Content Security Policy headers, and missing Sub Resource Integrity attributes.

Table 4.4.1 – Absence of Anti-CSRF Tokens

Vulnerability Identified	Absence of Anti-CSRF Tokens
Threat Category (S/T/R/I/D/E)	T – Tampering
Vulnerable URL/Parameters	http://127.0.0.1:8000/auth/login/?next=/
Severity	Medium

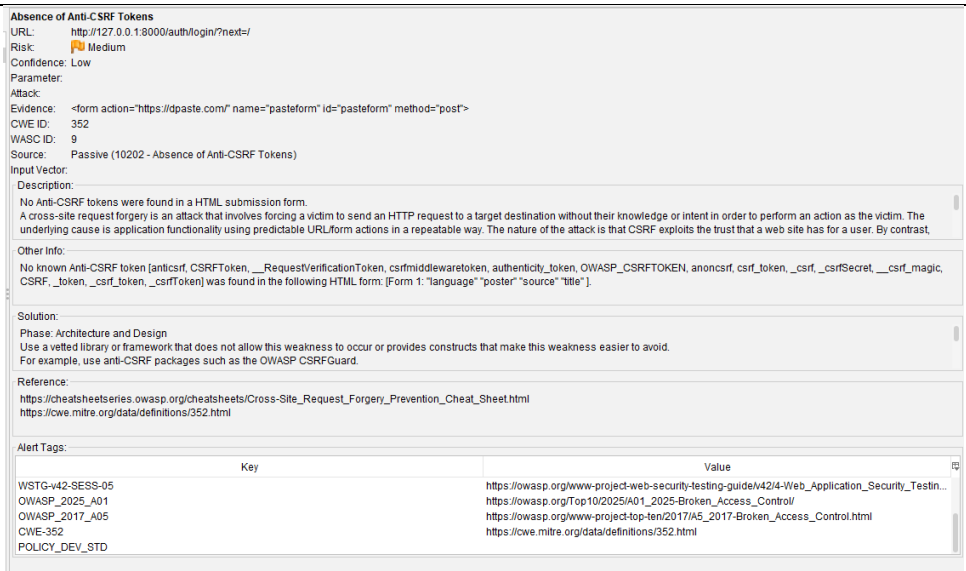
Steps Performed to Identify the Vulnerability	Step 1: Started the Django application locally. Step 2: Performed an automated scan using OWASP ZAP. Step 3: Reviewed the Alerts tab and identified forms without CSRF protection.
Screenshots	 The screenshot shows an alert titled 'Absence of Anti-CSRF Tokens' with a risk level of Medium. It details the URL, risk, confidence, and attack vector. The description explains that no Anti-CSRF tokens were found in a HTML submission form, which is a vulnerability as it allows for cross-site request forgery. The solution suggests using a vetted library or framework that prevents this weakness. Reference links to OWASP and CWE resources are provided. The alert tags section lists WSTG-v42-SESS-05, OWASP_2025_A01, OWASP_2017_A05, CWE-352, and POLICY_DEV_STD.
Faulty / Unsecure Coding	Forms did not contain a detectable Anti-CSRF token.
Remediation / Mitigation	Implement Django CSRF protection using {% csrf_token %} and ensure CsrfViewMiddleware is enabled.

Table 4.4.2 – Content Security Policy (CSP) Header Not Set

Vulnerability Identified	Content Security Policy (CSP) Header Not Set
Threat Category (S/T/R/I/D/E)	I – Information Disclosure
Vulnerable URL/Parameters	Application Response Headers
Severity	Medium
Steps Performed to Identify the Vulnerability	Step 1: Performed automated scanning using OWASP ZAP. Step 2: Inspected response headers. Step 3: Verified that no Content Security Policy header was configured.

Screenshots	Content Security Policy (CSP) Header Not Set URL: http://127.0.0.1:8000 Risk: Medium Confidence: High Parameter: Attack: Evidence: CWE ID: 693 WASC ID: 15 Source: Passive (10038 - Content Security Policy (CSP) Header Not Set) Alert Reference: 10038-1 Input Vector: Description: Content Security Policy (CSP) is an added layer of security that helps to detect and mitigate certain types of attacks, including Cross Site Scripting (XSS) and data injection attacks. These attacks are used for everything from data theft to site defacement or distribution of malware. CSP provides a set of standard HTTP headers that allow website owners to declare approved sources of content that browsers should be allowed to load on that page — covered types are JavaScript, CSS, HTML frames, fonts, images and embeddable objects such as Java applets, ActiveX, audio and video files. Other Info: Solution: Ensure that your web server, application server, load balancer, etc. is configured to set the Content-Security-Policy header. Reference: https://developer.mozilla.org/en-US/docs/Web/HTTP/Guides/CSP https://cheatsheetsseries.owasp.org/cheatsheets/Content_Security_Policy_Cheat_Sheet.html https://www.w3.org/TR/CSP/ Alert Tags: <table border="1"> <thead> <tr> <th>Key</th><th>Value</th></tr> </thead> <tbody> <tr> <td>POLICY_PENTEST</td><td></td></tr> <tr> <td>SYSTEMIC</td><td>https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic</td></tr> <tr> <td>CWE-693</td><td>https://cwe.mitre.org/data/definitions/693.html</td></tr> <tr> <td>OWASP_2017_A06</td><td>https://owasp.org/www-project-top-ten/2017/A6_2017-Security_Misconfiguration.html</td></tr> <tr> <td>OWASP_2025_A02</td><td>https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/</td></tr> </tbody> </table> Current Status	Key	Value	POLICY_PENTEST		SYSTEMIC	https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic	CWE-693	https://cwe.mitre.org/data/definitions/693.html	OWASP_2017_A06	https://owasp.org/www-project-top-ten/2017/A6_2017-Security_Misconfiguration.html	OWASP_2025_A02	https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/
Key	Value												
POLICY_PENTEST													
SYSTEMIC	https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic												
CWE-693	https://cwe.mitre.org/data/definitions/693.html												
OWASP_2017_A06	https://owasp.org/www-project-top-ten/2017/A6_2017-Security_Misconfiguration.html												
OWASP_2025_A02	https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/												
Faulty / Unsecure Coding	No Content Security Policy header was configured in HTTP responses.												
Remediation / Mitigation	Configure a Content Security Policy header to restrict trusted sources for scripts, styles, and other resources.												

Table 4.4.3 – Sub Resource Integrity (SRI) Attribute Missing

Vulnerability Identified	Sub Resource Integrity (SRI) Attribute Missing
Threat Category (S/T/R/I/D/E)	T – Tampering
Vulnerable URL/Parameters	External JavaScript and CSS resources loaded by the application
Severity	Low
Steps Performed to Identify the Vulnerability	Step 1: Started the Django application locally. Step 2: Performed an automated scan using OWASP ZAP. Step 3: Reviewed the Alerts tab and identified external resources without SRI attributes.

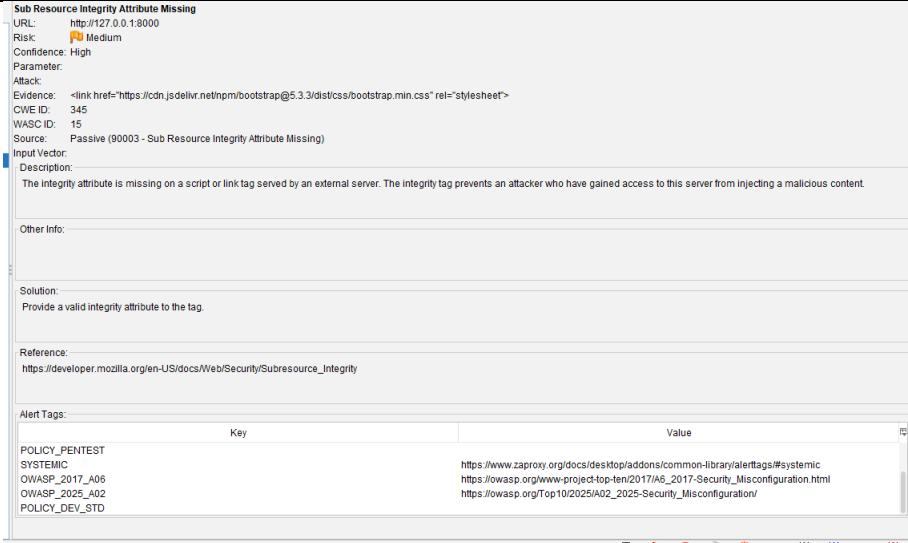
Screenshots	 Sub Resource Integrity Attribute Missing URL: http://127.0.0.1:8000 Risk: Medium Confidence: High Parameter: /link Evidence: <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css" rel="stylesheet"> CWE ID: 345 WASC ID: 15 Source: Passive (90003 - Sub Resource Integrity Attribute Missing) Input Vector: Description: The integrity attribute is missing on a script or link tag served by an external server. The integrity tag prevents an attacker who have gained access to this server from injecting a malicious content. Other Info: Solution: Provide a valid integrity attribute to the tag Reference: https://developer.mozilla.org/en-US/docs/Web/Security/Subresource_Integrity Alert Tags: <table border="1"> <thead> <tr> <th>Key</th> <th>Value</th> </tr> </thead> <tbody> <tr> <td>POLICY_PENTEST</td> <td>https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic</td> </tr> <tr> <td>SYSTEMIC</td> <td>https://owasp.org/www-project-top-ten/2017/1A6_2017-Security_Misconfiguration.html</td> </tr> <tr> <td>OWASP_2017_A06</td> <td>https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/</td> </tr> <tr> <td>OWASP_2025_A02</td> <td></td> </tr> <tr> <td>POLICY_DEV_STD</td> <td></td> </tr> </tbody> </table>	Key	Value	POLICY_PENTEST	https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic	SYSTEMIC	https://owasp.org/www-project-top-ten/2017/1A6_2017-Security_Misconfiguration.html	OWASP_2017_A06	https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/	OWASP_2025_A02		POLICY_DEV_STD	
Key	Value												
POLICY_PENTEST	https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#systemic												
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OWASP_2017_A06	https://owasp.org/Top10/2025/A02_2025-Security_Misconfiguration/												
OWASP_2025_A02													
POLICY_DEV_STD													
Faulty / Unsecure Coding	External scripts and stylesheets were loaded without integrity attributes, preventing browsers from verifying that the files had not been modified by an attacker.												
Remediation / Mitigation	Implement Sub Resource Integrity (SRI) by adding integrity and crossorigin attributes to externally loaded JavaScript and CSS resources. This ensures that browsers verify the authenticity and integrity of external files before executing them.												

Figure 32

Conclusion

The dynamic testing assessment successfully evaluated the running application and identified several medium and low-risk security findings. No critical vulnerabilities were detected during the automated scan. The findings provide recommendations that can be implemented to further improve the overall security of the Secure Task Management System.

4.5 Penetration Test Summary

A penetration testing assessment was conducted using multiple security testing techniques to evaluate the overall security of the Secure Task Management System. The assessment included manual code review, Software Component Analysis (SCA), static analysis, and dynamic testing. The purpose of these activities was to identify security weaknesses, verify the effectiveness of implemented security controls, and ensure compliance with secure coding practices.

Summary of Testing Activities

Test Category	Tool / Method Used	Result
---------------	--------------------	--------

Manual Code Review	Source Code Inspection	14/14 security controls passed
Software Component Analysis (SCA)	Snyk	Vulnerabilities identified in third-party dependencies
Static Analysis	Semgrep	51 findings detected for further review
Dynamic Testing	OWASP ZAP	Medium and low-risk findings identified
Fuzzy Testing	Manual Input Testing	Pass
Untrusted Input Testing	Form Validation Testing	Pass
SQL Injection Testing	Django ORM Validation	Pass
Cross-Site Scripting (XSS) Testing	Input Sanitization Testing	Pass
Authentication Testing	Login and Password Controls	Pass
Access Control Testing	RBAC and IDOR Verification	Pass
File Upload Testing	File Validation Controls	Pass

Overall Findings

Security Area	Status
Input Validation	Passed
Authentication Security	Passed
Authorization and Access Control	Passed
Password Security	Passed

File Upload Security	Passed
Audit Logging	Passed
Dependency Security	Improvement Required
Dynamic Security Controls	Improvement Required

Discussion

The penetration testing assessment showed that the Secure Task Management System implements several important security controls, including input validation, secure password hashing, role-based access control, audit logging, and secure file upload validation. Manual code review confirmed compliance with OWASP Top 10 secure coding practices, while static analysis and dynamic testing helped identify areas for further improvement.

The Software Component Analysis identified vulnerabilities within third-party dependencies, particularly outdated framework components. Dynamic testing using OWASP ZAP also revealed several medium and low-risk findings, including missing Anti-CSRF protection and missing Content Security Policy headers. Although these findings do not represent critical vulnerabilities, they should be addressed to further strengthen the security posture of the application.

Conclusion

Based on the penetration testing activities performed, the Secure Task Management System demonstrates a good level of security and follows secure coding practices. No critical vulnerabilities were identified within the application's custom source code. Future improvements should focus on updating vulnerable dependencies, implementing additional security headers, and enhancing protection against common web application attacks to further improve overall system security.

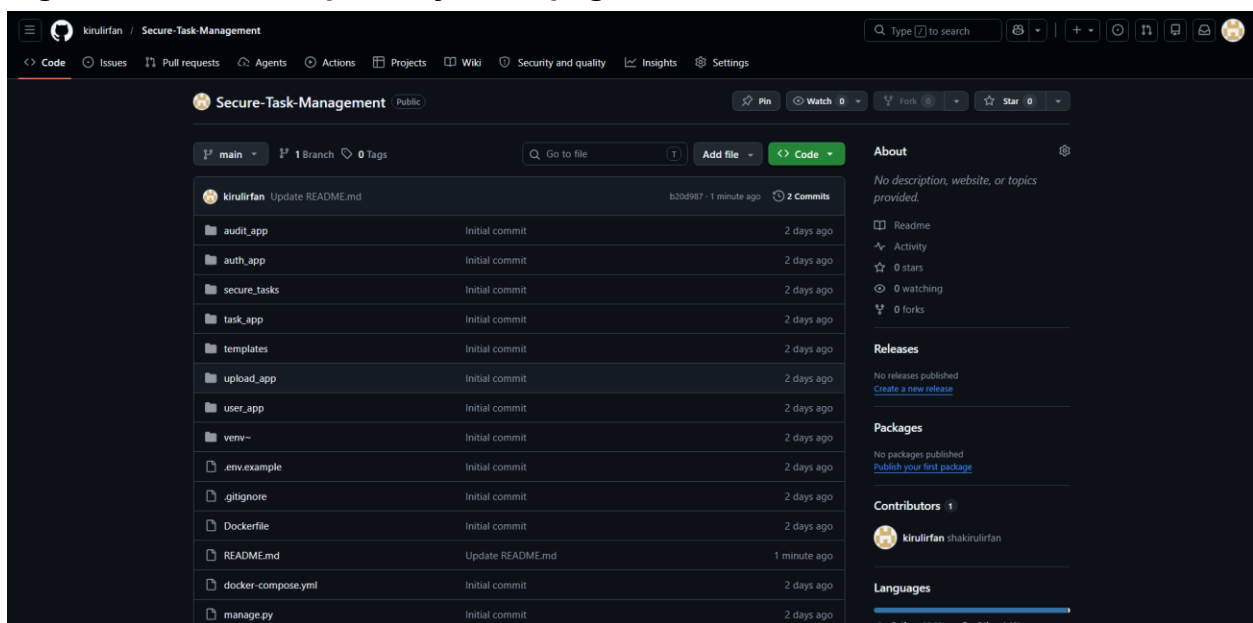
5.1 GitHub Repository Details

GitHub was used as the version control platform for the Secure Task Management System. The repository was used to store source code, manage project files, track development progress, and maintain version history throughout the software development lifecycle.

Item	Details
------	---------

Repository Name	Secure-Task-Management
Repository Link	https://github.com/kirulirfan/Secure-Task-Management
GitHub Username	kirulirfan
Version Control Platform	GitHub
Repository Purpose	Source Code Management and Version Control

Figure 34: GitHub Repository Homepage



Description:

The screenshot shows the GitHub repository used to manage the Secure Task Management System source code and project documentation.

5.3 Repository Structure

The project repository is organized into multiple modules to improve maintainability, readability, and security management.

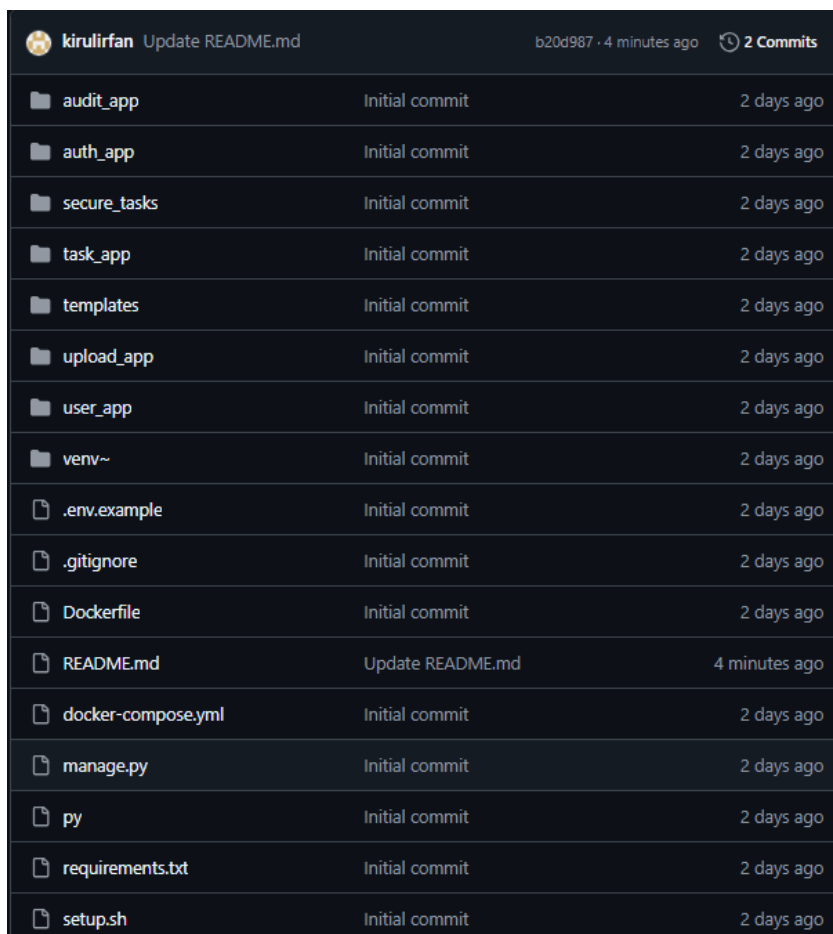
Repository Structure

```
/audit_app
/auth_app
/secure_tasks
/task_app
```

/templates
/upload_app
/user_app
/venv

.env.example
.gitignore
Dockerfile
README.md
docker-compose.yml
manage.py
requirements.txt
setup.sh

Figure 35: Repository Structure



 kirulirfan Update README.md	b20d987 · 4 minutes ago	 2 Commits
 audit_app	Initial commit	2 days ago
 auth_app	Initial commit	2 days ago
 secure_tasks	Initial commit	2 days ago
 task_app	Initial commit	2 days ago
 templates	Initial commit	2 days ago
 upload_app	Initial commit	2 days ago
 user_app	Initial commit	2 days ago
 venv~	Initial commit	2 days ago
 .env.example	Initial commit	2 days ago
 .gitignore	Initial commit	2 days ago
 Dockerfile	Initial commit	2 days ago
 README.md	Update README.md	4 minutes ago
 docker-compose.yml	Initial commit	2 days ago
 manage.py	Initial commit	2 days ago
 py	Initial commit	2 days ago
 requirements.txt	Initial commit	2 days ago
 setup.sh	Initial commit	2 days ago

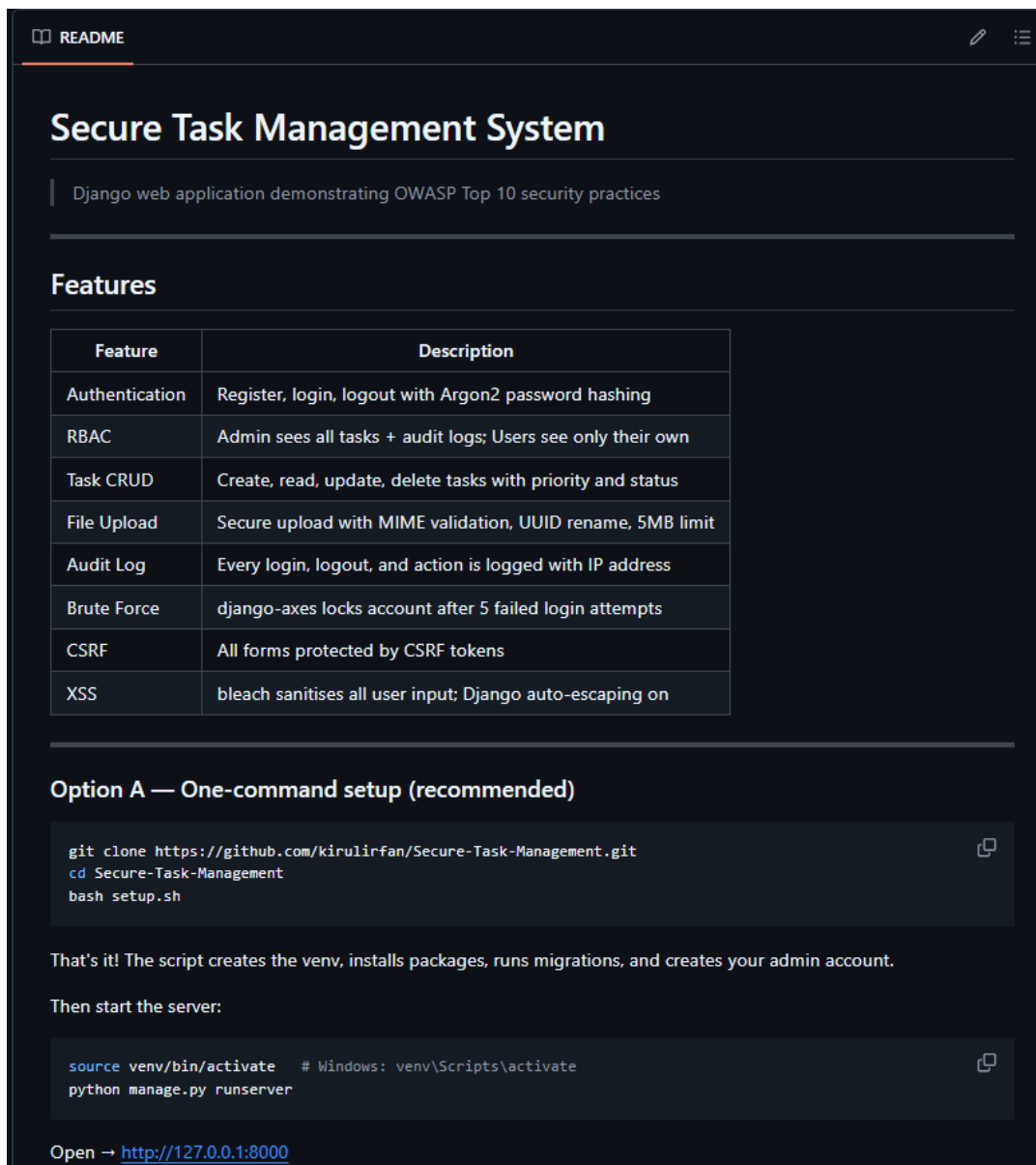
Description:

The screenshot shows the directory structure and project files stored within the GitHub repository.

5.4 README Documentation

A README document was prepared to provide installation instructions, project information, security features, dependencies, usage guidelines, and system screenshots. This documentation helps users and developers understand the system configuration and deployment process.

Figure 36: README Documentation



The screenshot shows a README file for a project titled "Secure Task Management System". The header includes a "README" tab and a description: "Django web application demonstrating OWASP Top 10 security practices". Below the header is a "Features" section containing a table with 8 rows. The table has two columns: "Feature" and "Description". The features listed are Authentication, RBAC, Task CRUD, File Upload, Audit Log, Brute Force, CSRF, and XSS. Below the table is a section titled "Option A — One-command setup (recommended)" which contains a code block with three lines of terminal commands: `git clone https://github.com/kirulirfan/Secure-Task-Management.git`, `cd Secure-Task-Management`, and `bash setup.sh`. Below the code block is a paragraph explaining that the script creates the venv, installs packages, runs migrations, and creates an admin account. It then says "Then start the server:" followed by another code block with two lines: `source venv/bin/activate` (with a note for Windows: `venv\Scripts\activate`) and `python manage.py runserver`. At the bottom, it says "Open →" followed by the URL <http://127.0.0.1:8000>.

README

Secure Task Management System

Django web application demonstrating OWASP Top 10 security practices

Features

Feature	Description
Authentication	Register, login, logout with Argon2 password hashing
RBAC	Admin sees all tasks + audit logs; Users see only their own
Task CRUD	Create, read, update, delete tasks with priority and status
File Upload	Secure upload with MIME validation, UUID rename, 5MB limit
Audit Log	Every login, logout, and action is logged with IP address
Brute Force	django-axes locks account after 5 failed login attempts
CSRF	All forms protected by CSRF tokens
XSS	bleach sanitises all user input; Django auto-escaping on

Option A — One-command setup (recommended)

```
git clone https://github.com/kirulirfan/Secure-Task-Management.git
cd Secure-Task-Management
bash setup.sh
```

That's it! The script creates the venv, installs packages, runs migrations, and creates your admin account.

Then start the server:

```
source venv/bin/activate # Windows: venv\Scripts\activate
python manage.py runserver
```

Open → <http://127.0.0.1:8000>

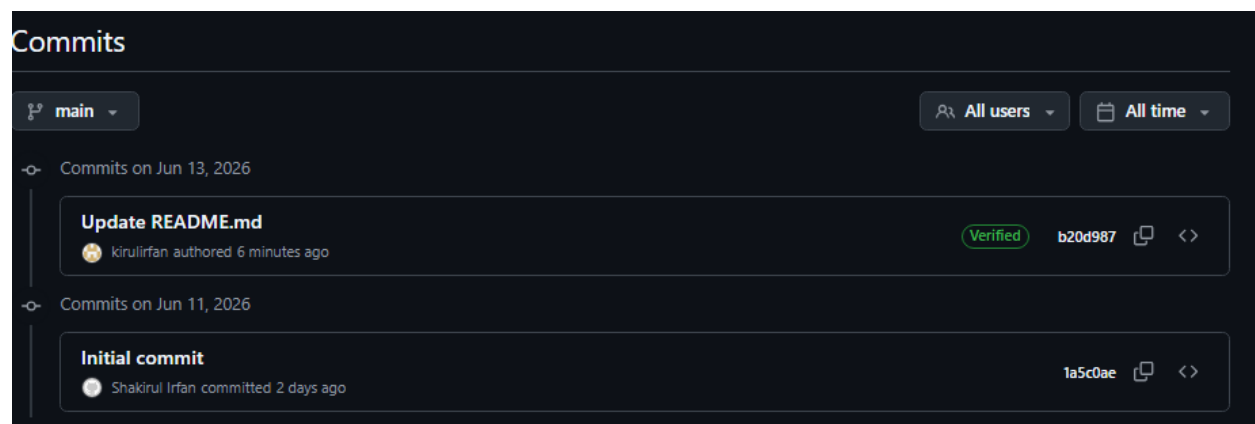
Description:

The screenshot shows the README documentation containing project information, installation instructions, security features, dependencies, and system screenshots.

5.5 GitHub Commit History

GitHub commit history was used to track project development activities, feature implementation, bug fixes, and security improvements throughout the software development lifecycle. Version control ensures that all project changes are recorded and can be reviewed when necessary.

Figure 37: GitHub Commit History



Description:

The screenshot displays the commit history recorded in GitHub throughout the development of the Secure Task Management System.

Conclusion

GitHub successfully supported the development and management of the Secure Task Management System throughout the project lifecycle. The repository provided a centralized platform for source code storage, version control, documentation management, and progress tracking. Features such as commit history and repository organization helped maintain project consistency and allowed development activities to be recorded systematically.

6.0 Results and Discussion

The security assessment conducted on the Secure Task Management System included manual code review, Software Component Analysis (SCA), static analysis, and dynamic testing. The objective of these assessments was to evaluate the effectiveness of the implemented security controls and identify potential security weaknesses within the application.

The manual code review confirmed that the application follows secure coding practices based on OWASP Top 10 guidelines. Security controls such as input validation, secure password hashing, role-based access control, audit logging, file upload validation, and brute-force protection were successfully implemented and verified through source code inspection.

Software Component Analysis using Snyk identified vulnerabilities within several third-party dependencies. These findings highlight the importance of maintaining updated software libraries and regularly monitoring dependency security throughout the software development lifecycle.

Static analysis using Semgrep successfully scanned the application source code and identified 51 findings that require developer review. While many findings represent potential risks rather than confirmed vulnerabilities, the results provide valuable insight into areas that may require further attention.

Dynamic testing using OWASP ZAP identified several medium and low-risk findings, including the absence of Anti-CSRF tokens, missing Content Security Policy (CSP) headers, and missing Sub Resource Integrity (SRI) attributes. No critical vulnerabilities were discovered during the assessment.

Overall, the results demonstrate that the Secure Task Management System incorporates multiple layers of security controls and follows secure development practices. The identified findings provide opportunities for future improvement and further enhancement of the system's security posture.

7.0 Conclusion

This project successfully developed a Secure Task Management System that incorporates secure coding practices based on the OWASP Top 10 security guidelines. The system provides essential task management functionalities while implementing multiple security controls to protect user data and reduce common web application vulnerabilities.

Several security mechanisms were implemented throughout the development process, including secure authentication, strong password policies, role-based access control, audit logging, file upload validation, input sanitization, and brute-force protection. These controls were designed to improve the confidentiality, integrity, and availability of the system.

To evaluate the effectiveness of the implemented security measures, multiple security testing techniques were conducted. These included manual code review, Software Component Analysis (SCA) using Snyk, static analysis using Semgrep, and dynamic testing using OWASP ZAP. The testing results confirmed that the application follows secure software development practices and effectively mitigates many common security threats.

Although several medium and low-risk findings were identified during automated testing, no critical vulnerabilities were discovered within the application's custom source code. The identified findings mainly involve security hardening and dependency management improvements.

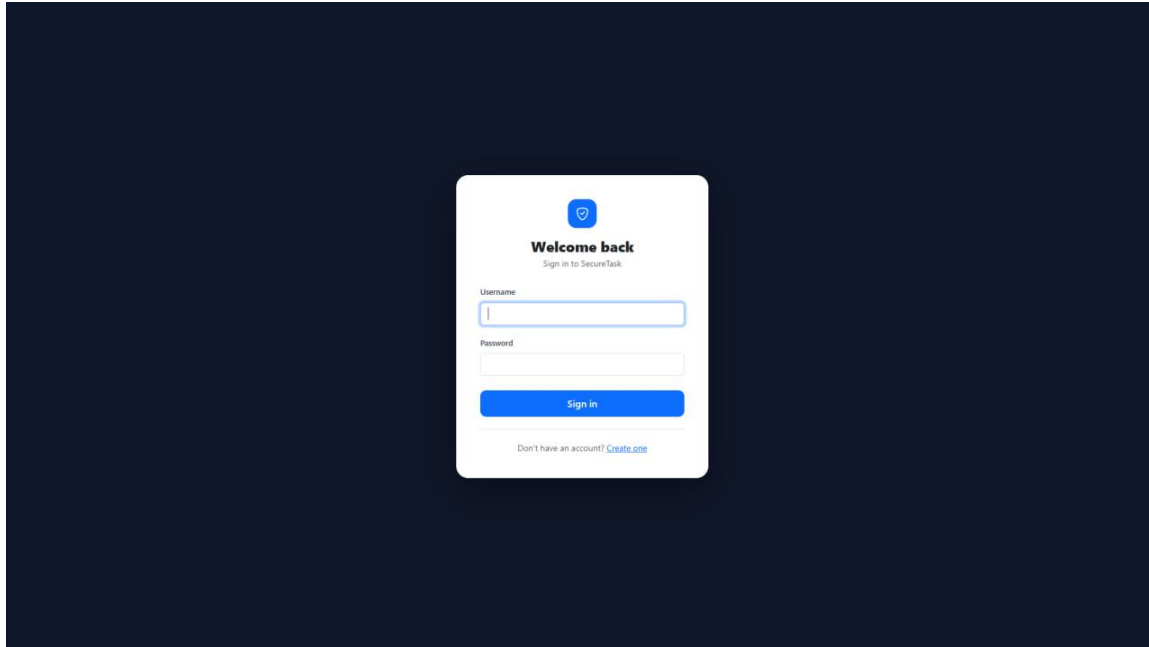
In conclusion, the Secure Task Management System achieved its objectives of demonstrating secure software development practices while providing a functional and secure web-based task management solution. Future improvements should focus on updating vulnerable dependencies, implementing additional security headers, and conducting regular security assessments to maintain a strong security posture.

8.0 References

9.0 Appendix

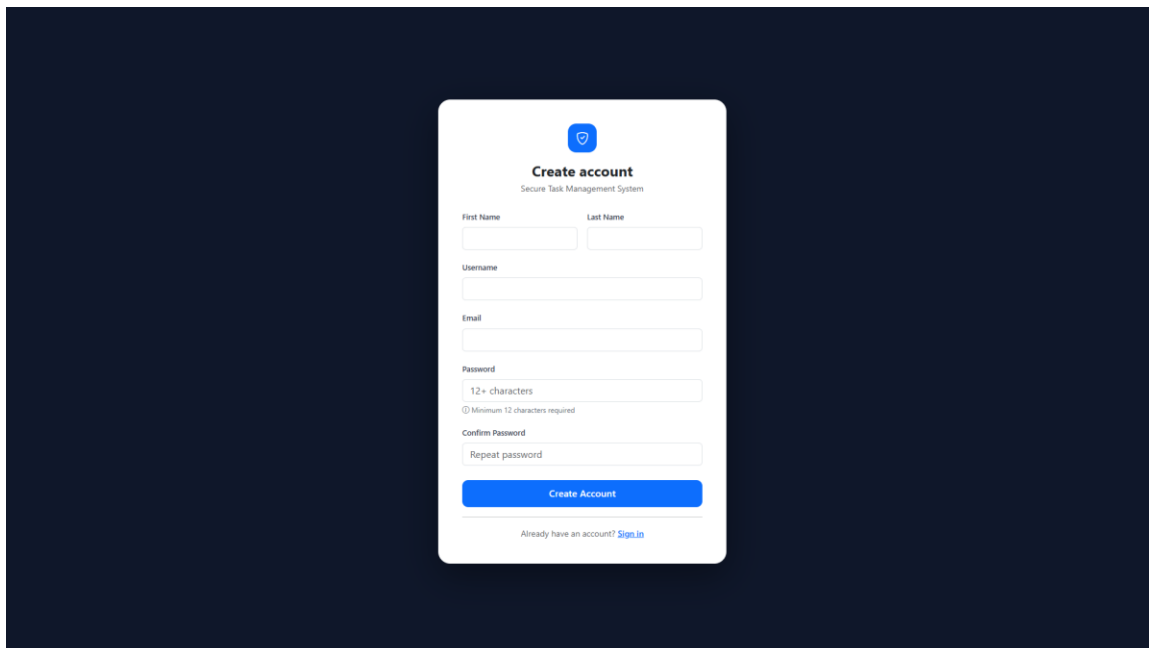
Appendix A – System Screenshots

Figure A1: Login Page



The login page features a dark blue background with a white login card in the center. The card has a blue shield icon at the top, followed by the text "Welcome back" and "Sign in to SecureTask". Below this are input fields for "Username" and "Password", a blue "Sign in" button, and a link "Don't have an account? [Create one](#)".

Figure A2: Registration Page



The registration page features a dark blue background with a white registration card in the center. The card has a blue shield icon at the top, followed by the text "Create account" and "Secure Task Management System". Below this are input fields for "First Name", "Last Name", "Username", "Email", "Password" (with a "12+ characters" requirement and a "Minimum 12 characters required" note), and "Confirm Password" (with a "Repeat password" note). A blue "Create Account" button is at the bottom, with a link "Already have an account? [Sign In](#)" below it.

Figure A3: Dashboard

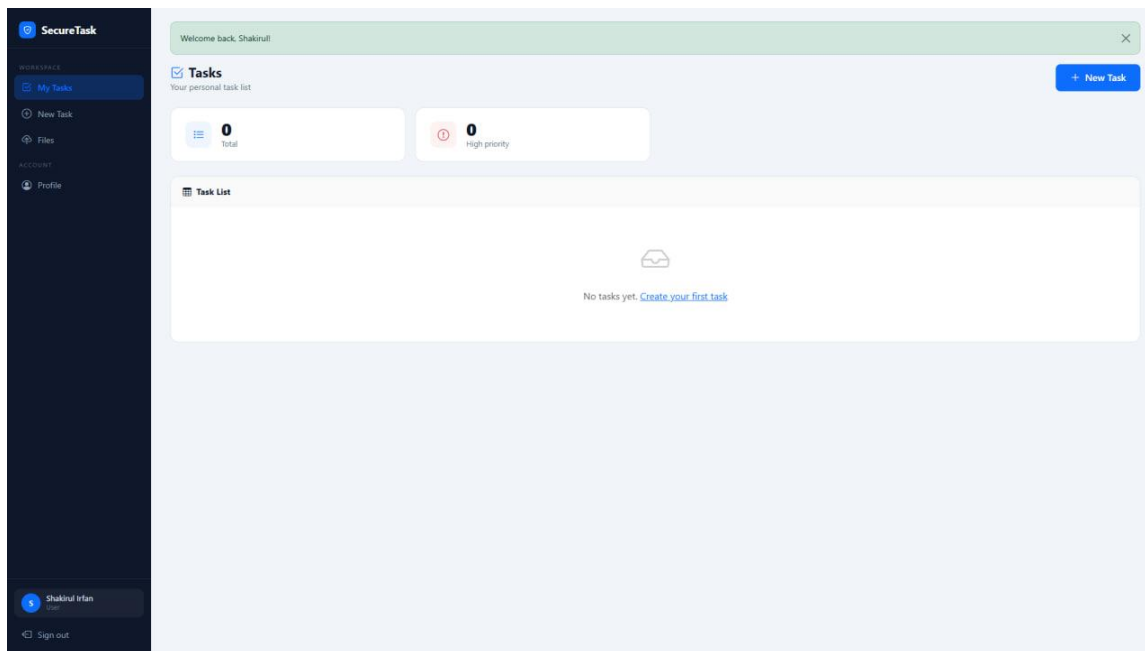


Figure A4: Task Management Page

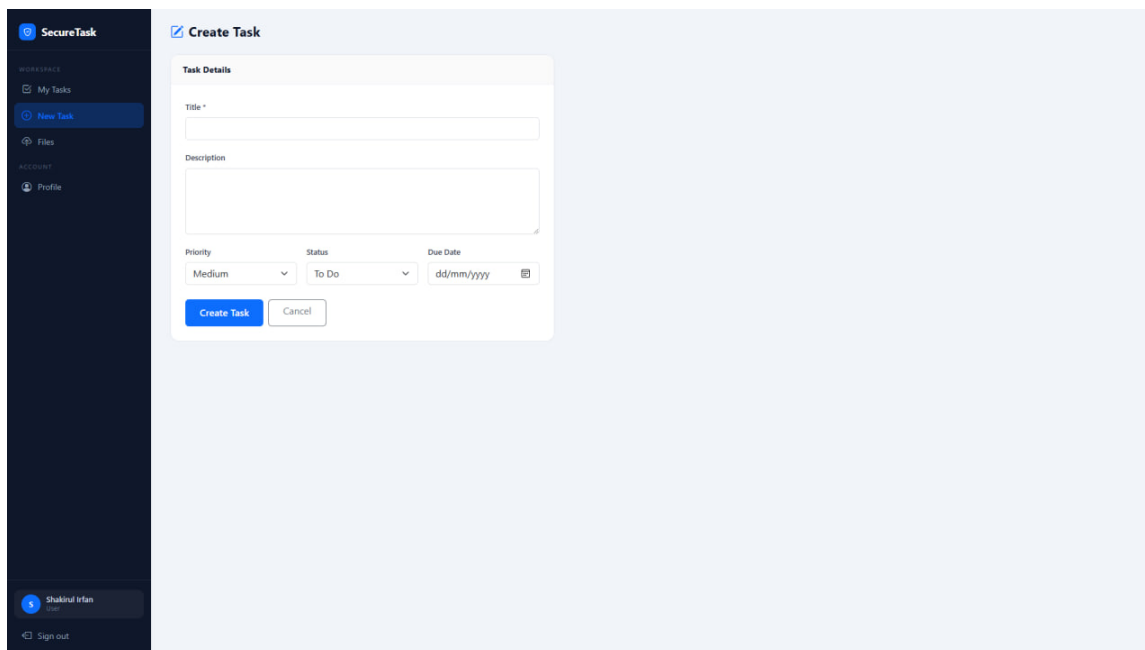


Figure A5: File Upload Page

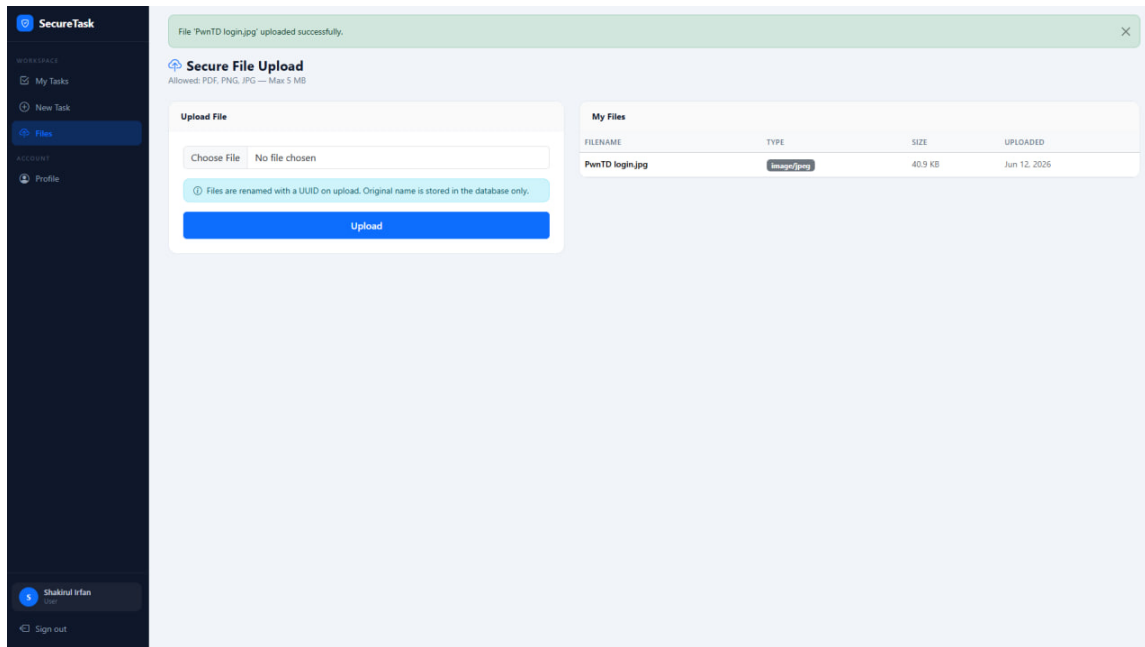


Figure A6: Audit Log Page

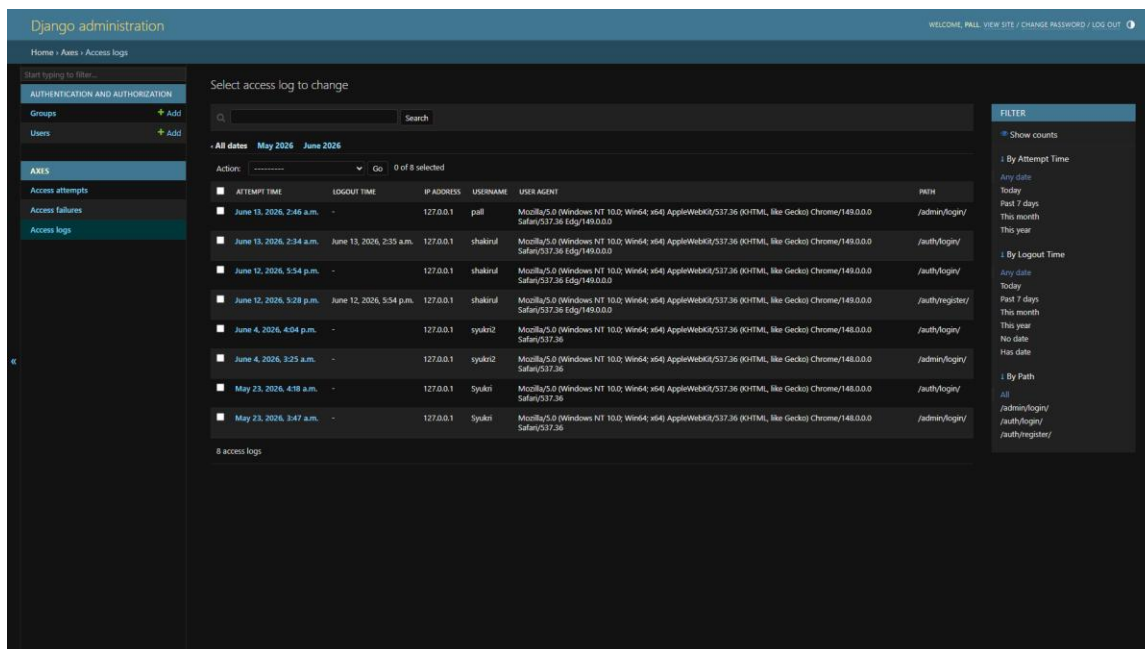


Figure A7: User Profile Page

SecureTask

My Profile

Shakirul Irfan

shakeyzzzz123@gmail.com

Joined Jun 2026

1 tasks

Edit Profile

First Name

Shakirul

Last Name

Irfan

Bio

Max 500 characters. HTML is stripped for security.

Save Changes

Shakirul Irfan

Sign out

127.0.0.1:8000/profile/